

Scalable ML Assignment

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1 Log Mining and Analysis

Table 1: Total requests and unique hostnames by UK sector.

Sector	Total Requests	Unique Hosts
Academic	25009	1022
Company	22225	1116
Government	195	15

For the `.ac.uk` hosts, the institution domain is extracted by first splitting each hostname using the full stop delimiter. For example, the Sheffield hostname, `www.shef.ac.uk`, is split into `["www", "shef", "ac", "uk"]`. The final three labels are concatenated together, resulting in `shef.ac.uk`, which represents the institution domain. Different hosts from the same institution (`cs.shef.ac.uk`, `www.shef.ac.uk`, etc.) are grouped together within the Sheffield domain.

Listing 1 shows the Spark implementation of the method described above.

Listing 1: Institution domain extraction and aggregation.

```
academic_hosts = logs.filter(col("host").endsWith(".ac.uk"))

split_host = split(col("host"), "\\.")

academic_domains = academic_hosts.select(
    col("host"),
    concat_ws(".",
        element_at(split_host, -3),
        element_at(split_host, -2),
        element_at(split_host, -1)
    ).alias("institution")
)

institution_counts = academic_domains.groupBy("institution") \
    .count() \
    .withColumnRenamed("count", "total_requests")
```

The total number of requests from the University of Sheffield domain is 623. The seven institutions with more requests than Sheffield, and their respective total requests, are shown below:

Table 2: UK academic institutions with more requests than the University of Sheffield.

Institution	Total Requests
<code>hensa.ac.uk</code>	4257
<code>rl.ac.uk</code>	1158
<code>ucl.ac.uk</code>	1036
<code>man.ac.uk</code>	921
<code>ic.ac.uk</code>	851
<code>soton.ac.uk</code>	808
<code>bham.ac.uk</code>	629

The nine most active UK (`.co.uk`) domains and their respective total requests are displayed in Figure 1. A bar chart is chosen to compare request counts across discrete company categories. It clearly

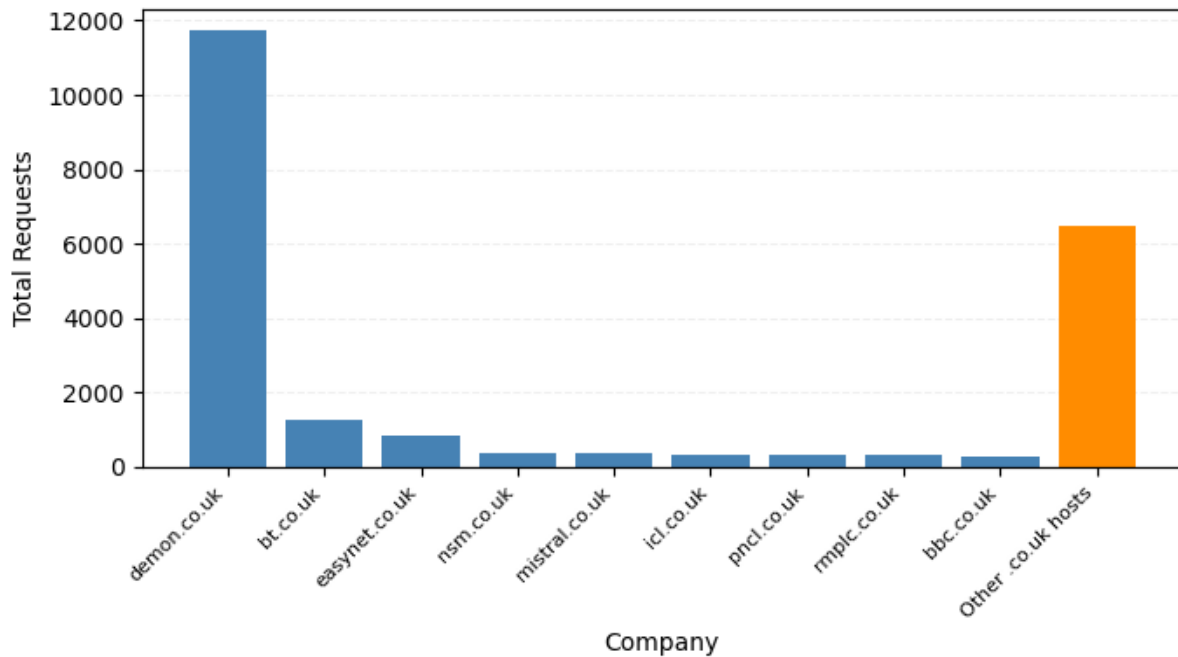


Figure 1: Total requests from the nine most active UK company domains, compared with the combined total for all remaining UK companies.

shows the top nine companies alongside all remaining companies combined, with contrasting colours used to highlight the comparison and improve readability.

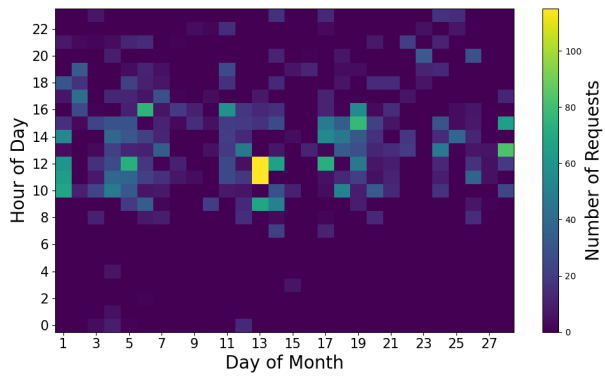
The most active academic institution is `hensa.ac.uk`, while the most active company domain is `demon.co.uk` Figure 2 shows the hourly request patterns for these domains and all `.gov.uk` hosts combined.

The heatmaps show that access activity is typically concentrated during daytime and evening hours. The `demon.co.uk` has the densest activity overall, whereas the combined `.gov.uk` requests are much more infrequent.

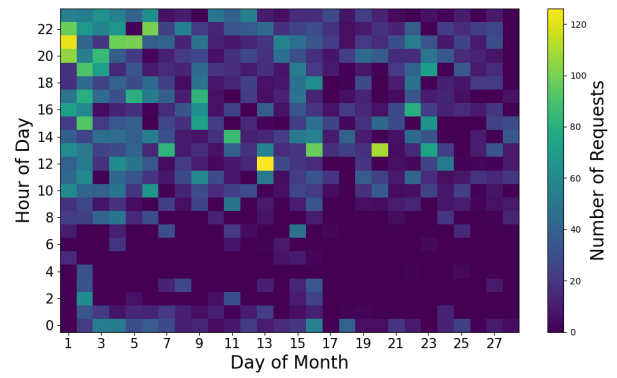
2 Scalable Logistic Regression and Generalised Linear Models

3 Scalable Decision Trees, Random Forests, and Gradient Boosting

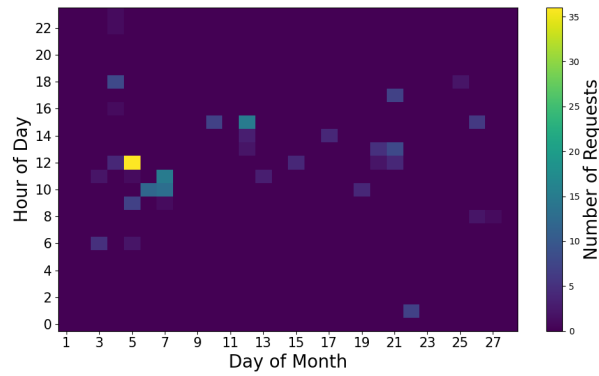
4 Movie Recommendations and Cluster Analysis



(a) `hensa.ac.uk`.



(b) `demon.co.uk`.



(c) All `.gov.uk` domains.

Figure 2: Hourly access patterns across days 1–28. Colour intensity represents the number of requests.